

41. Diabetic Kidney Disease · assets 1-9 / 27

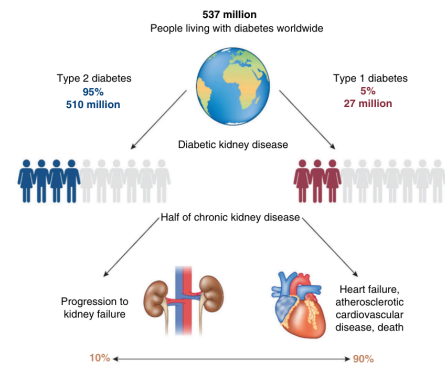


Fig. 41.1 Worldwide prevalence of diabetes.²² The worldwide prevalence of diabetes was estimated at 537 million people in the year 2021. Approximately 40% of people with type 2 diabetes (95% of all diabetes cases) and 30% of those with type 1 diabetes (5% of all diabetes cases) develop diabetic kidney disease. Half of all cases of CKD are attributable to diabetes. Most people with diabetic kidney disease (90%) die, mainly of heart failure or atherosclerotic cardiovascular events, with only a minority (10%) surviving to reach kidney failure.

Fig_41_1

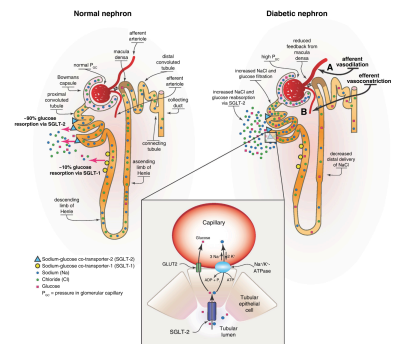


Fig. 41.4 Normal and diabetic nephron with altered hemodynamics. (A) Effluent vasodilation is promoted by hyperglycemia, hyperinsulinemia, elevated levels of circulating angiotensin, COX-2 prostanoids, and reduction of tubule glomerular feedback. Tubuloglomerular feedback is an intrinsic autoregulatory mechanism that helps maintain the rate of glomerular filtration rate. Because of increased reabsorption of glucose and sodium via SGLTs in diabetes, sodium chloride delivery to macula densa cells of juxtaglomerular apparatus is decreased, resulting in lower production of adenosine and consequent natriuretic vasodilation of afferent arteriole. (B) Effluent vasoconstriction is promoted by high local angiotensin II levels, endothelin 1, reactive oxygen species, and thromboxane A_2 .

Fig_41_4

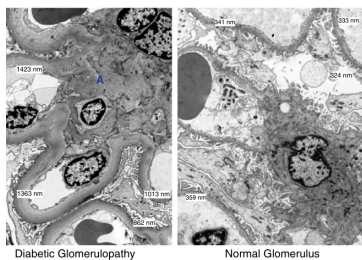


Fig. 41.7 Electron microscope images of structural changes in diabetic kidney disease.³⁴⁸ Structural changes in diabetic glomerulopathy found with electron microscopy. *A* indicates marked expansion of the mesangium. *B* indicates marked diffuse thickening of capillary basement membranes (to three times the normal thickness in this case). *C* indicates segmental effacement of the visceral epithelial foot processes. Original magnification, $\times 3500$.

Fig_41_7

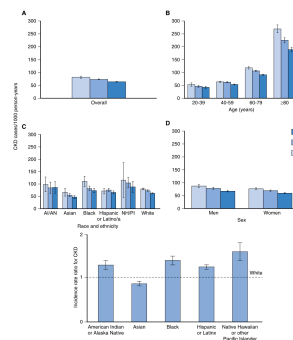


Fig. 41-2 Incidence rate ratios for chronic kidney disease among patients with diabetes.²⁷ Panel 1: (A) Chronic kidney disease with 95% confidence intervals (CIs) in diabetes over time. Overall CKD incidence standardized to the 2010 U.S. population by age, sex and ethnicity, and sex. (B) Age-specific, adjusted for sex, race, and ethnicity. (C) Race and ethnicity-specific for sex and age. (D) Sex-specific, adjusted for age, race, and ethnicity. *AI/AN*, American Indian/Alaskan Native; *NPY*, Native Hawaiian. Panel 2: Incidence rate ratios for CKD among patients with diabetes, stratified according to race and ethnic group, between 2002. The dashed line represents the incidence rate ratio among White patients (reference group). The analysis was adjusted for

Fig_41_2

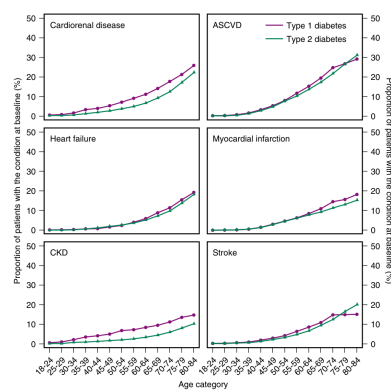


Fig. 41.5 Rates of cardiovascular and kidney disease in type 1 and type 2 diabetes.³³³ Age-stratified baseline prevalence of chronic kidney disease (CKD), heart failure (HF), cardiorenal disease (HF or CKD), stroke, myocardial infarction (MI), and atherosclerotic cardiovascular disease (ASCVD) (MI or stroke).

Fig_41_5

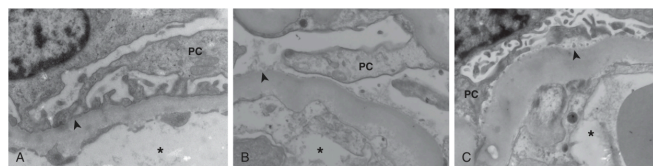
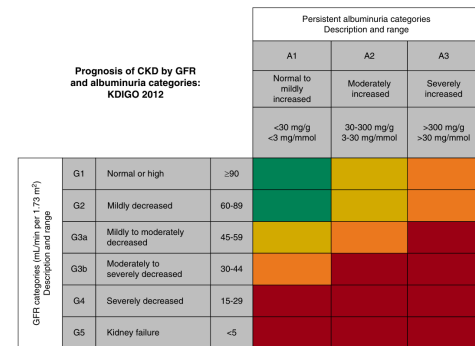


Fig. 41.8 Detachment of podocytes (PC) from the glomerular basement membrane in diabetic kidney disease (DKD).³⁶⁴ (A) Glomerular basement membrane covered by intact foot processes in mild DKD. With advancement of DKD, some areas of glomerular basement membranes with complete (B), or partial (C) detachment of podocytes are observable. *Capillary lumen. Arrowhead shows the podocyte aspect of the glomerular basement membrane.

Fig_41_8



Green: low risk (if no other markers of kidney disease, no CKD); yellow: moderately increased risk; orange: high risk; red, very high risk.

Fig. 41.3 Categorization and prognosis of chronic kidney disease.³

Fig_41_3

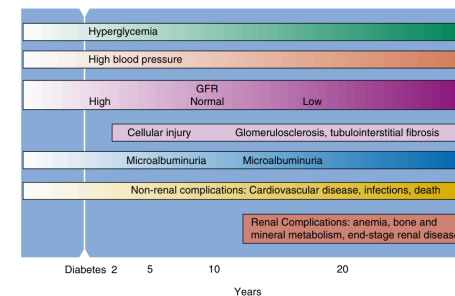


Fig. 41.6 Conceptual model of the natural history of diabetic kidney disease.³⁴⁵ Duration of diabetes, in years, is presented on the horizontal axis. Timeline is well characterized for type 1 diabetes mellitus; for type 2 diabetes mellitus, timeline may depart from the illustration due to the variable timing of the onset of hyperglycemia. Kidney complications: anemia, bone and mineral metabolism, retinopathy, and neuropathy.

Fig_41_6

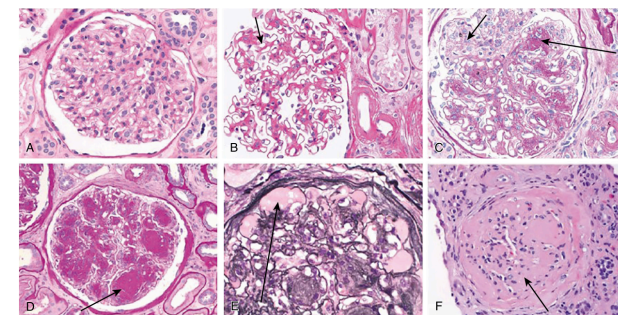


Fig. 41.9 Glomerular changes in diabetic kidney disease.³⁴⁵ Changes in glomerular histology in diabetic glomerulopathy. (A) Normal glomerulus. (B) Diffuse mesangial expansion with mesangial cell proliferation. (C) Prominent mesangial expansion with early nodularity and mesangiolysis. (D) Accumulation of mesangial matrix forming Kimmelstiel-Wilson nodules. (E) Dilatation of capillaries forming microaneurysms, with subintimal hyaline (plasmatic infarction). (F) Obsolescent glomerulus. A–D and F were stained with period acid–Schiff stain, and E was stained with Jones stain. Original magnification, $\times 400$.

Fig_41_9

Table 41.2. Biomarkers of Diabetic Kidney Disease		
Class	Biomarkers	Method of detection
Currently used	eGFR	Radioisotope or creatinine
	Albuminuria	Urine
	Urea nitrogen	Blood
	Urea nitrogen	Blood
Oxidative stress	Endothelial dysfunction	Blood
	Endothelial dysfunction	Blood
	Endothelial dysfunction	Blood
	Endothelial dysfunction	Blood
Fibrosis	ECM	Blood
	ECM	Blood
	ECM	Blood
	ECM	Blood
Glomerular damage	ECM	Blood
	ECM	Blood
	ECM	Blood
	ECM	Blood
Tubular damage	ECM	Blood
	ECM	Blood
	ECM	Blood
	ECM	Blood
Inflammation	ECM	Blood
	ECM	Blood
	ECM	Blood
	ECM	Blood
Microalbuminuria	ECM	Blood
	ECM	Blood
	ECM	Blood
	ECM	Blood

Table_41_2

Table 41.5 KDIGO Practice Points for use of Renin-Angiotensin System Blockers in Patients with Diabetic Kidney Disease ²²
• ACE inhibitor or ARB therapy may be considered in patients with diabetes, albuminuria, and normal blood pressure. • After starting or increasing the dose of an ACE inhibitor or ARB, monitor for changes in blood pressure, serum creatinine, and serum potassium within 2-4 weeks. • ACE inhibitor/ARB therapy should be continued unless serum creatinine increases by more than 30% within 4 weeks of drug initiation or an increase in dose. • Women receiving an ACE inhibitor or ARB should be advised about appropriate contraception and should be advised to discontinue therapy if they are considering pregnancy or become pregnant. • Hyperkalemia associated with ACE inhibitor/ARB use can often be managed by interventions to reduce serum potassium levels without decreasing the dose or stopping the ACE inhibitor or ARB immediately. • Reduce the dose or discontinue ACE inhibitor or ARB therapy with either symptomatic hypotension or uncontrolled hyperkalemia despite alternative medical treatment approaches, or to reduce uricemic symptoms in the setting of kidney failure. • Do not use dual renin-angiotensin system blockade. ACE, Angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; KDIGO, Kidney Disease Improving Global Outcomes.

Table_41_5

Table 41.8 Summary of Finerenone Outcome Trials ^{63,628}		
Trial	FIDELIO-DKD (n = 6734)	FIGARO-DKD (n = 7437)
Treatment	Finerenone vs. placebo	Finerenone vs. placebo
Mean participant age (years)	66	64
Key inclusion criteria	• T2D • eGFR 25 to <60 mL/min/1.73 m ² and UACR 30 to <300 mg/g, or • eGFR 25 to <75 mL/min/1.73 m ² and UACR 300 to <500 mg/g • Treated with RAS inhibitor at maximum tolerated dose	• T2D • eGFR 25 to 90 mL/min/1.73 m ² and UACR 30 to <300 mg/g, or • eGFR <60 mL/min/1.73 m ² and UACR 300 to <500 mg/g • Treated with RAS inhibitor at maximum tolerated dose
Mean baseline A1C (%)	7.7	7.7
Median follow-up (years)	2.6	3.4
Primary outcome	HR (95% CI)	HR (95% CI)
HR (95% CI)	Kidney failure, ≥40% decline in eGFR, or renal death 0.82 (0.73-0.93)	CV death, nonfatal MI, nonfatal stroke, or hospitalization for HF 0.87 (0.76-0.98)
Key secondary outcomes		
Key secondary composite; HR (95% CI)	CV death, nonfatal MI, nonfatal stroke, or hospitalization for HF 0.86 (0.75-0.99)	Kidney failure, ≥40% decline in eGFR, or renal death 0.87 (0.76-1.01)

A1C, Glycated hemoglobin A1c; CI, confidence interval; CV, cardiovascular; eGFR, estimated glomerular filtration rate; HF, heart failure; HR, hazard ratio; MI, myocardial infarction; RAS, renin-angiotensin system; T2D, type 2 diabetes mellitus; UACR, urinary albumin-to-creatinine ratio.

Table_41_8

Table 41.3 Classification of Diabetic Kidney Disease Glomerular Lesions According to Tervaert et al ²⁷¹		
Class	Description	Inclusion criteria
I	Mild or nonspecific LM changes and EM-proven GBM thickening	Biopsy does not meet any of the criteria mentioned below for class II, III, or IV GBM >395 nm in female and >430 nm in male individuals 9 years of age and older
IIa	Mild mesangial expansion	Biopsy does not meet criteria for class III or IV Mild mesangial expansion in >25% of the observed mesangium
IIb	Severe mesangial expansion	Biopsy does not meet criteria for class III or IV Severe mesangial expansion in >25% of the observed mesangium
III	Nodular sclerosis (Kimmelstiel-Wilson lesion)	Biopsy does not meet criteria for class IV At least one convincing Kimmelstiel-Wilson lesion
IV	Advanced diabetic glomerulosclerosis	Global glomerular sclerosis in >50% of glomeruli Lesions from classes I through III

^aOn the basis of direct measurement of GBM width by EM, these individual cutoff levels may be considered indicative when other GBM measurements are used.
EM, Electron microscopy; GBM, glomerular basement membrane; LM, light microscopy.

Table_41_3

Table 41.6 SGLT2 Inhibitor Kidney Outcome Trials Summary ⁶⁷⁹⁻⁶⁸⁰			
Study	CRENDENCE (N = 4401)	DAPA-CKD (N = 4304)	EMPA-KIDNEY (N = 6609)
Agent	Canagliflozin	Dapagliflozin	Dapagliflozin
Median follow-up	2.6 years	2.4 years	2.0 years
Diabetes status	T2D	T2D or non-T2D CKD	T2D or non-T2D CKD
Kidney-related enrollment criteria	• eGFR ≥30 to <90 mL/min/1.73 m ² • UACR: ≥300 to <5000 mg/g	• eGFR ≥25 to <75 mL/min/1.73 m ² • UACR: ≥200 to 5000 mg/g	• eGFR ≥20 to <45 mL/min/1.73 m ² OR • eGFR ≥45 to <90 mL/min/1.73 m ² with UACR ≥200 mg/g
Mean baseline eGFR	56 mL/min/1.73 m ²	43 mL/min/1.73 m ²	37.5 mL/min/1.73 m ²
Median baseline UACR	927 mg/g	949 mg/g	412 mg/g
Kidney outcome(s)	Primary Outcome • ESKD (dialysis, transplantation, or sustained eGFR <15 mL/min/1.73m ²), doubling of SCR, or death from renal causes HR: 0.70 (0.59-0.82)	Primary Outcome • ≥50% decrease in eGFR, ESKD, or death from renal or cardiovascular causes HR: 0.61 (0.51-0.72)	Primary Outcome • ≥40% decrease in eGFR, ESKD, sustained decline in eGFR to <10 mL/min/1.73m ² or cardiovascular death HR: 0.72 (0.64-0.82)

eGFR, Estimated glomerular filtration rate; ESKD, end-stage kidney disease; HR, hazard ratio; SCR, serum creatinine; T2D, type 2 diabetes; UACR, urinary albumin-to-creatinine ratio.

Table_41_6

Table 41.9 Factors that May Guide Decisions on Individualized A1C Targets in Diabetic Kidney Disease ^{30,31,363,369}		
Factor/Consideration	Supports less stringent A1C target (e.g., <8.0%)	Supports more stringent A1C target (e.g., <6.5% or 7.0%)
CKD severity	Advanced-stage CKD (e.g., CKD G5)	Early-stage CKD (e.g., CKD G1)
Macrovascular complications	Present/Severe	Absent/Minor
Comorbidity burden	High	Low
Life expectancy	Short	Long
Hypoglycemia risk	High	Low
Hypoglycemia awareness	Impaired	Present
Resources/Support	Scarce	Available
Patient motivation/preference for treatment intensity	Low motivation	Highly motivated

Adapted from El Sayed NA, Aleppo G, Aroda VR, et al. American Diabetes Association. Standards of Care in Diabetes—2023. Diabetes Care. 2023;46(suppl 1):S1–S291 and Kidney Disease: Improving Global Outcomes Diabetes Work Group. KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease. Kidney Int. 2022;102:S1–S127.

Table_41_9

Table 41.4 Classification of Diabetic Kidney Disease Interstitial and Vascular Lesions According to Tervaert et al ²⁷¹		
Lesion	Criteria	Score
Interstitial lesions		
IFTA	No IFTA	0
	<25%	1
	25%-50%	2
	>50%	3
	Absent	0
Vascular lesions		
Arterial hyaline	Absent	0
	At least 1 area of arteriolar hyaline	1
	More than 1 area of arteriolar hyaline	2
Presence of large vessels		
Arteriosclerosis (score worst artery)	No intimal thickening	Yes/no
	Intimal thickening less than thickness of media	1
	Intimal thickening greater than thickness of media	2

IFTA, Interstitial fibrosis and tubular atrophy.

Table_41_4

Table 41.7 Key Monitoring and Risk Mitigation Strategies for Agents Recommended in Patients with type 2 Diabetes and Chronic Kidney Disease ^{60,683}		
Medication class	Consideration	Monitoring and/or risk mitigation strategies
SGLT2 inhibitors	• Volume depletion	• Proactive dose reduction of diuretics in patients at high risk for hypovolemia
	• Diabetic ketoacidosis (DKA)	• Hold SGLT2 inhibitors during illness
	• Genital mycotic infections	• Educate about signs/symptoms to facilitate early recognition
	• Hypoglycemia	• Monitor blood or urine ketones for very high risk
	• Metformin-associated lactic acidosis	• Institute a sick day protocol
Metformin		• Maintain at least low-dose insulin in insulin-requiring individuals
		• Counsel on genital hygiene (e.g., regular bathing and wearing clean undergarments)
		• Adjust background glucose-lowering agents (e.g., insulin and/or sulfonylureas) as appropriate
		• Monitor eGFR with increasing frequency as eGFR falls <60 mL/min/1.73 m ²
		• Adjust metformin dose as appropriate per eGFR
		• Consider dose reduction in the presence of conditions that predispose patients to hypoperfusion and hypoxemia if eGFR 45-59 mL/min/1.73 m ²
		• Institute a sick day protocol
		• Monitor patients for vitamin B ₁₂ deficiency when treated with metformin for >4 years
GLP1 receptor agonists	• Nausea/vomiting/diarrhea	• Start at lowest recommended dose and titrate slowly
	• Hypoglycemia	• Educate on potential GI intolerance and importance of communicating severe symptoms to care team
	• Hyperkalemia	• Adjust background glucose-lowering agents (e.g., insulin and/or sulfonylureas) as appropriate
Nonsteroidal MRAs		• Monitor serum potassium prior to initiation and during treatment

eGFR, Estimated glomerular filtration rate; GI, gastrointestinal; GLP1, glucagon-like peptide-1; SGLT2, sodium-glucose cotransporter-2.

Table_41_7

Table 41.10 Moderate- and High-Intensity Statin Therapy⁶⁴¹

Moderate intensity (lowers LDL by 30%-49%)	High intensity (lowers LDL by ≥50%)
• Atorvastatin 10-20 mg	• Atorvastatin 40-80 mg
• Rosuvastatin 5-10 mg	• Rosuvastatin 20-40 mg
• Simvastatin 20-40 mg	
• Pravastatin 40-80 mg	
• Lovastatin 40 mg	
• Fluvastatin XL 80 mg	
• Pitavastatin 1-4 mg	

All dosages are once-daily dosing.
XL, Extended release.

Table_41_10